EndoSAM: Pelvic Organ Segmentation for Endometriosis Diagnosis using Deep Learning and Foundation Models

Antony Gerold Arockiasamy – 1404714
*ENGG*6302 – Medical Image Processing*
University of Guelph

Segmenting pelvic organs in MRI is important for diagnosing endometriosis and planning surgery. I evaluated multiple deep learning methods on the UT-EndoMRI dataset (124 subjects) to segment uterus, ovary, and endometrioma. While working with the data, I found that 52% of subjects had spatial misalignment between masks and images due to multi-sequence MRI acquisition. Fixing this improved MedSAM Dice from 0.356 to 0.842 for uterus, which was bigger than any model change. MedSAM with decoder fine-tuning gave the best results (uterus 0.857, ovary 0.783, endometrioma 0.744). Multi-class U-Net failed for ovary (Dice 0.000) due to class imbalance, but a binary model with Focal Tversky Loss reached 0.298, matching the published RAovSeg baseline of 0.290. I also report the first endometrioma segmentation result at 0.556 Dice when detected.

***Index Terms*—medical image segmentation, endometriosis, MRI, MedSAM, U-Net**

I. Introduction

Endometriosis affects about 190 million women worldwide [1]. It is a condition where tissue similar to the uterine lining grows outside the uterus, usually on or near the ovaries. Around 17 to 44 percent of patients develop endometriomas, which are blood-filled cysts on the ovary [2]. Getting a proper diagnosis still takes 7 to 10 years on average [3].

MRI is the recommended way to image endometriosis [4]. Knowing the exact boundaries of the uterus, ovary, and any cysts helps surgeons preserve healthy tissue during operations. But drawing these boundaries manually on every MRI slice takes a lot of time and different radiologists often disagree on where the edges are.

The UT-EndoMRI dataset [5] is the first public benchmark for this task. It has 124 subjects with expert annotations. The authors proposed RAovSeg which gets 0.290 Dice for ovary using a two-stage pipeline. To my knowledge, no published work applies foundation models like MedSAM [6] to this data, and endometrioma segmentation has not been attempted.

This work makes four contributions: (1) identification and correction of a mask alignment problem affecting more than half the dataset, (2) the first evaluation of MedSAM on pelvic endometriosis MRI, (3) binary organ-specific models that match the published baseline with a simpler pipeline, and (4) the first endometrioma segmentation result.

II. Literature Review

A. U-Net and Variants

U-Net [7] is the standard architecture for medical image segmentation. The encoder compresses the image through convolution and pooling while the decoder brings it back to full resolution through upsampling. Skip connections between matching encoder and decoder levels pass high-resolution details that would otherwise get lost. Without these connections the output boundaries would be blurry.

Attention U-Net [8] adds gates that help the model focus on areas more likely to contain the target organ. Roy et al. [18] proposed SCSE which recalibrates both where the model looks (spatial) and what features it pays attention to (channel). nnU-Net [9] is a self-configuring framework that automatically picks the right preprocessing, architecture, and training settings for each dataset. It has become one of the strongest baselines in the field. Transfer learning from ImageNet-pretrained encoders like EfficientNet [19] is common practice, especially when training data is limited.

B. Loss Functions for Imbalanced Data

Standard cross-entropy treats all pixels equally, which is a problem when the target organ is less than 1% of the image. The model can get high accuracy just by predicting background everywhere. Dice loss [10] directly measures overlap and is less affected by this imbalance. Focal loss [11] puts more weight on pixels that are hard to classify. Tversky loss [12] lets you separately control how much to penalize false positives versus false negatives. When missing the organ is worse than predicting a few extra pixels, you can set the weights accordingly.

C. Foundation Models

SAM [13] changed how segmentation works. You give it an image plus a prompt like a bounding box and it produces a mask. It was trained on over 1 billion masks from natural images. MedSAM [6] adapted this for medical imaging by fine-tuning on 1.5 million image-mask pairs from 10 different types of medical scans. It has three parts: an image encoder (ViT-B, 90M parameters), a prompt encoder (6K parameters), and a mask decoder (4M parameters).

The main limitation is that someone still needs to draw a bounding box. Gaillochet et al. [14] replaced the prompt encoder with a small learnable module that generates prompts automatically from the image, needing only 10 training samples. Zhou et al. [15] proposed MedSAM-U which uses uncertainty estimation to refine prompts automatically. Li et al. [16] showed that training with randomly perturbed boxes makes the model more robust when the box is not perfect. MedSAM-2 [17] extends everything to 3D volumes. BiomedParse [20] takes a different approach using text prompts instead of spatial prompts.

D. Pelvic Organ Segmentation

Pelvic MRI segmentation is particularly hard. The ovary is only 0.3 to 2.5 percent of the image pixels. The UT-EndoMRI dataset [5] has subjects from 15 imaging sites with 9 different scanners, so there is a lot of variation. RAovSeg [5] uses a ResNet-18 classifier to find which slices contain ovary, then an Attention U-Net to segment them. Their most important finding is an ablation: without the classifier, ovary Dice drops from 0.290 to 0.013. This proves the main issue is class imbalance, not model capacity. nnU-Net gets 0.272. Neither paper reports endometrioma segmentation. Other related work includes uterine fibroid segmentation using U-Net on MRI [21].

III. Methods

A. Dataset

The UT-EndoMRI dataset [5] has 124 subjects from two hospitals. D1 (51 subjects) from Memorial Hermann across 15 sites with 9 scanners, with T2 and T1FS sequences, 3 raters per structure. D2 (73 subjects) from Texas Children's with four sequences (T2, T2FS, T1FS, T1) and 1 rater. Each structure (uterus, ovary, endometrioma/cyst) is annotated as a separate binary NIfTI file. Data is split 70/15/15 by subject with seed 42.

B. Evaluation Metric

I use the Dice Similarity Coefficient as the primary metric:

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|} \tag{1}$$

where P is the predicted mask and G is the ground truth. It ranges from 0 (no overlap) to 1 (perfect match). I did not compute boundary metrics like Hausdorff distance in this study, though they would be useful for evaluating surgical planning accuracy.

C. Mask Alignment Problem and Fix

When I first ran MedSAM on this data, I got 0.356 Dice for uterus. That seemed way too low so I started looking at individual subjects. I found that the masks and images did not always have matching dimensions.

The reason is that MRI scanners capture multiple sequences of the same patient in one session. T2 might be 560 by 473 pixels while T1FS is 700 by 569. The radiologist picked whichever sequence showed the organ best for drawing the annotation, but a standard loader just grabs the first image file it finds. So you end up with a 700 by 569 mask on a 560 by 473 image and everything is misaligned.

My fix checks the width and height of each mask file and finds the image file with matching dimensions. I do this per-mask because sometimes different structures in the same subject were annotated on different sequences. After this fix, 96 out of 124 subjects are usable versus 59 before. MedSAM Dice for uterus went from 0.356 to 0.842, an improvement of +0.486 from data quality alone.

D. Inter-Rater Agreement

D1 has 3 raters per structure so I computed pairwise Dice between all combinations. I noticed that 8 out of 35 comparisons had raters who annotated on different sequences, which gives artificially low Dice because of the alignment issue. After excluding those, the corrected inter-rater Dice is 0.605 ± 0.386 for uterus and 0.345 ± 0.377 for ovary. The ovary value is lower than liver (0.92), kidney (0.87), prostate (0.83), and pancreas (0.78). Radiologists themselves struggle with ovary boundaries in endometriosis.

E. Preprocessing

All images are normalized by clipping to the 1st and 99th percentile of foreground pixels and scaling to 0–1. For U-Net, input is 256 by 256 with 3 channels (target slice plus one above and one below for context). For MedSAM, input is 1024 by 1024 as required by the ViT-B encoder.

F. MedSAM

MedSAM takes an image and a bounding box prompt and outputs a segmentation mask. The image encoder (ViT-B) splits the 1024 by 1024 image into 4096 patches of 16 by 16 pixels each and uses self-attention so every patch can see every other patch, unlike convolutions which only look at local neighborhoods. I used the official Zenodo weights (medsam_vit_b.pth). I tested HuggingFace community weights and they gave much worse results (0.30–0.50 versus 0.85+).

Oracle Box: Tightest rectangle around ground truth plus 5–10 pixel margin. This is the best possible prompt.

Simulated Manual Box: I modeled what would happen if a radiologist drew the box by hand. The noise level depends on inter-rater agreement:

$$\sigma = d_{\text{bbox}} \times 0.15 \times (1 - D_{\text{inter-rater}}) \tag{2}$$

Organs where radiologists disagree more get noisier boxes. I generate 5 noisy boxes per slice and average the predictions.

Decoder Fine-Tuning: I froze the encoder (90M params) and prompt encoder (6K params) and only trained the decoder (4M params). This prevents overfitting since I only have about 80 representative samples. I pre-computed all encoder embeddings once to save time. The loss combines Binary Cross-Entropy and Dice loss:

$$\mathcal{L} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}} \tag{3}$$

I used AdamW with learning rate 10^{-4} and weight decay 10^{-4}, gradient clipping at 1.0, for 10 epochs. Training data:

1110 slices from 99 subjects. Test: 234 slices from 25 held-out subjects.

G. U-Net

Baseline and Attention: I used an EfficientNet-B0 [19] encoder (5.3M parameters) with a U-Net decoder. The output has 4 classes through softmax:

$$\mathrm{softmax}(z_c) = \frac{e^{z_c}}{\sum_{j=1}^{4} e^{z_j}} \tag{4}$$

This forces all four probabilities to add up to 1 at each pixel, which becomes a problem when one class dominates. The attention version adds SCSE modules [18] after each decoder block. I tried attention because in my previous work on retinal OCT imaging, spatial attention helped find thin retinal layers. I wanted to see if it would help with small pelvic structures. Both models used cross-entropy loss with Adam at LR=10^{-3} for 15 epochs on CPU.

Binary Models: Multi-class U-Net got Dice near zero for ovary and endometrioma on validation. After reading the RAovSeg ablation [5] I understood why: the softmax competition with 95%+ background pixels leaves no gradient for minority classes. Running multi-class on test confirmed this empirically (Dice 0.000 for ovary), which motivated the binary approach. I trained separate 2-class U-Net models for each organ, using only slices that contain the target organ. The Focal Tversky Loss is:

$$\mathrm{TI} = \frac{\mathrm{TP}}{\mathrm{TP} + 0.8 \cdot \mathrm{FN} + 0.2 \cdot \mathrm{FP}} \tag{5}$$

$$\mathcal{L}_{\mathrm{FTL}} = 0.7 \cdot (1 - \mathrm{TI})^{1.33} + 0.3 \cdot \mathcal{L}_{\mathrm{CE}} \tag{6}$$

The 0.8 on false negatives means missing an organ pixel hurts 4 times more than a false alarm. I used LR=3×10^{-4} after my first try at 10^{-3} caused the training to collapse at epoch 7.

H. Auto-Prompting Pipeline

To make segmentation fully automatic, I tried extracting bounding boxes from U-Net and binary model predictions and feeding them to the MedSAM decoder. I tested both multi-class and binary models as box generators.

IV. Results

A. Data Quality

Table I shows the effect of the alignment fix. Going from 0.356 to 0.842 for uterus just by loading the right image was bigger than any model change in this study.

TABLE I
IMPACT OF ALIGNMENT CORRECTION

	Before	**After**
Usable subjects	59	96
MedSAM oracle (uterus)	0.356	0.842
Inter-rater (uterus)	0.327	0.605

B. MedSAM

Table II shows the three prompting strategies. Zero-shot with oracle boxes already works well. Simulated manual boxes drop only 4 to 8 percent. Decoder fine-tuning gives the best numbers.

TABLE II
MEDSAM RESULTS (DICE)

Prompt	**Uterus**	**Ovary**	**Endo**
Oracle (304 sl.)	0.842	0.744	0.696
Simulated (304 sl.)	0.805	0.683	0.641
Decoder FT (234 sl.)	**0.857**	**0.783**	**0.744**

C. U-Net

Table III shows the U-Net results on test data. Baseline gets decent uterus but zero for ovary and endo. Attention actually did worse for uterus on test, probably overfitting from the extra parameters, but helped endo appear at 0.225. Binary models solve the ovary problem: 0.298 matches the published RAovSeg (0.290). Binary endometrioma reaches 0.556 when the organ is detected, with an overall mean of 0.389 across all test slices including those where the model predicted nothing.

TABLE III
U-NET RESULTS (DICE, TEST)

Method	**Uterus**	**Ovary**	**Endo**
Baseline U-Net	0.656	0.000	0.000
Attention U-Net	0.557	0.000	0.225
Binary Ovary	–	0.298	–
Binary Endo	–	–	0.556*

*When detected. Overall mean: 0.389 (70 test slices).

Fig. 1 shows what happens at the pixel level. In a typical slice, ovary is about 5% of pixels. Softmax makes all 4 classes compete and background wins every time. In the binary model the competition is removed and the organ shows up.

D. Auto-Prompting

Table IV shows the auto-box results. MedSAM does improve upon the U-Net boxes (uterus went from 0.170 to 0.392) but ovary detection is the bottleneck. Multi-class U-Net only finds ovary in 3 out of 83 slices. Binary models detect it in 63 out of 83 but the resulting Dice of 0.266 is far below oracle-prompted performance of 0.783, because the decoder was trained with perfect boxes and does not handle rough boxes well.

TABLE IV
AUTO-PROMPTING PIPELINE (DICE)

Box Source	**Uterus**	**Ovary**	**Endo**
Multi-class U-Net	0.392	0.093	0.275
Binary models	–	0.266	0.362

Fig. 1. Why multi-class fails for ovary. Top-left: ground truth with 47 background, 14 uterus, and 3 ovary pixels. Top-right: multi-class U-Net predicts all 3 ovary pixels as background, giving Dice = 0.000. Bottom-left: binary model trained on only ovary-containing slices with 2 classes instead of 4. Bottom-right: binary finds 4 of 5 ovary pixels, Dice = 0.800.

TABLE V
COMPLETE RESULTS (DICE)

Method	Ut.	Ov.	En.	Type
RAovSeg [5]	–	0.290	–	Published
nnU-Net [5]	–	0.272	–	Published
Inter-rater	0.605	0.345	–	Measured
Baseline U-Net	0.656	0.000	0.000	Test
Attention U-Net	0.557	0.000	0.225	Test
Binary Ovary	–	0.298	–	Test
Binary Endo	–	–	0.556	Test
MedSAM Oracle	0.842	0.744	0.696	304 sl.
MedSAM Simulated	0.805	0.683	0.641	304 sl.
MedSAM Dec. FT	**0.857**	**0.783**	**0.744**	234 sl.
Auto-box (U-Net)	0.392	0.093	0.275	Test
Auto-box (Binary)	–	0.266	0.362	Test

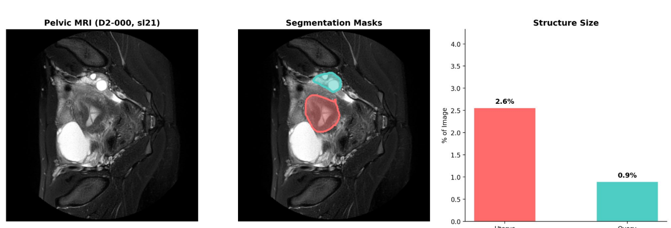

Fig. 2. MRI slice from subject D2-056 with organ overlays. Uterus (red) occupies 2.4% of the image while ovary (teal) is only 0.9%, illustrating the extreme class imbalance that causes multi-class U-Net to fail.

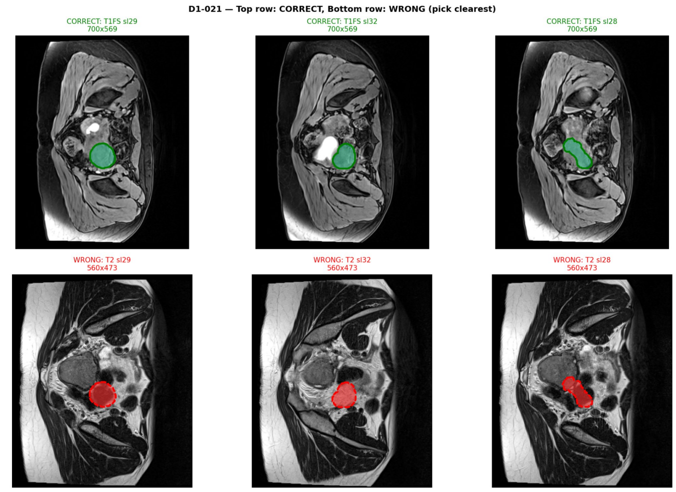

Fig. 3. Mask misalignment in subject D1-021 shown across three slices. Top row: correct alignment on T1FS (700 by 569 pixels). Bottom row: same mask overlaid on T2 (560 by 473 pixels), showing spatial shift that degrades Dice from 0.85+ to 0.35.

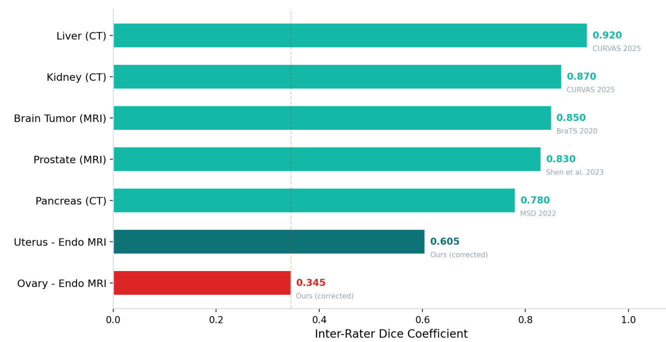

Fig. 4. Corrected inter-rater Dice across medical segmentation tasks. Values for other organs are from published benchmarks (sources labeled on bars). Ovary in endometriosis MRI at 0.345 is the lowest, though direct comparison is approximate since each task uses different datasets and annotation protocols.

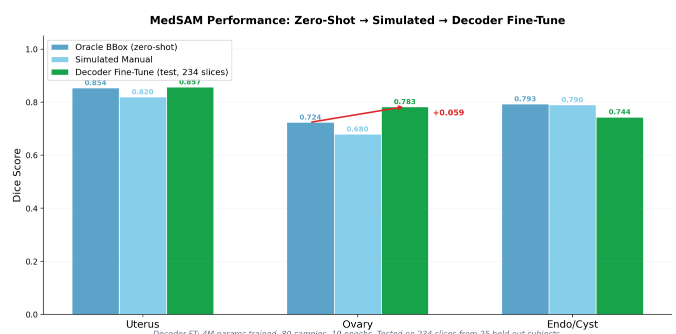

Fig. 5. MedSAM performance across prompting strategies. Decoder fine-tuning (green) improves over zero-shot oracle (blue) by +0.015 for uterus, +0.039 for ovary, and +0.048 for endometrioma. Simulated manual boxes (light blue) show only 4–8% drop from oracle.

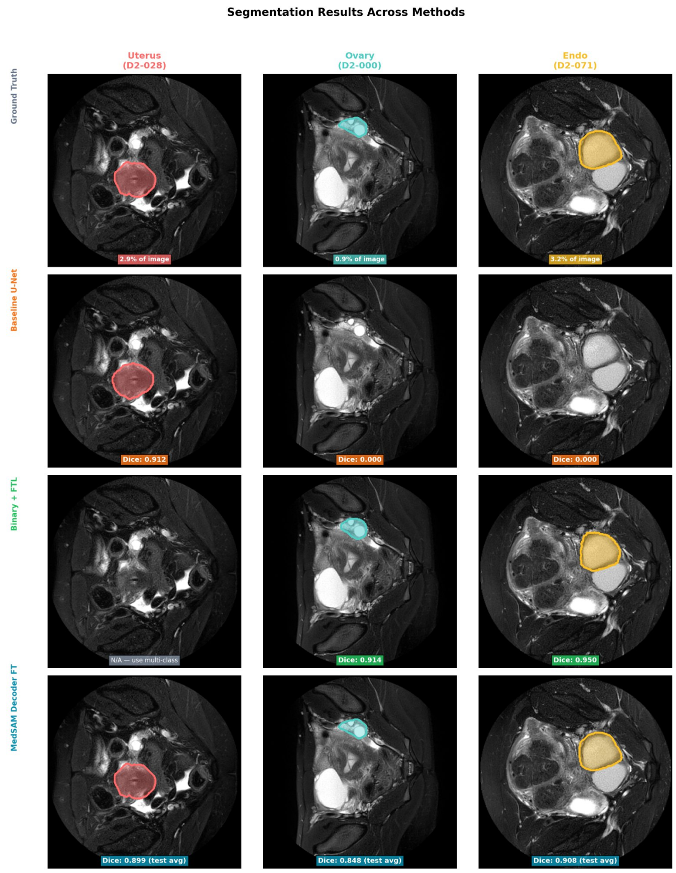

Fig. 6. Qualitative comparison on test subjects D2-028 (uterus), D2-000 (ovary), and D2-071 (endometrioma). Row 1: ground truth. Row 2: baseline U-Net (ovary column is empty, Dice 0.000). Row 3: binary model finds the organ. Row 4: MedSAM decoder gives the tightest boundaries.

E. Complete Comparison

Table V puts all results together.

V. Discussion

A. Data Quality

The biggest result in this project was that fixing the alignment gave +0.486 Dice. That is more than any model change. The problem happens because radiologists annotate on whatever sequence shows the organ best, but loaders assume a fixed sequence. My per-mask matching algorithm is simple but anyone using this dataset needs it.

B. Foundation Models

MedSAM got 0.842 for uterus without any training on pelvic data. It was trained on 1.5 million other medical images and the features still transfer. Decoder fine-tuning with only 4M trainable parameters and about 80 samples pushed it to 0.857 for uterus and 0.783 for ovary. The frozen encoder already knows medical image features so the decoder just needs to learn what pelvic organs look like specifically.

The simulated box experiment shows 4 to 8 percent drop from oracle. That is encouraging for clinical use since a radiologist drawing a rough box would get almost the same result as a perfect one.

C. Class Imbalance

Multi-class U-Net getting 0.000 for ovary on test confirms the theoretical prediction: with ovary at 0.5% of pixels, standard cross-entropy cannot produce meaningful gradients for minority classes. Softmax (Eq. 4) forces probabilities to sum to 1 across 4 classes, and the 95%+ background pixels dominate the gradient. This empirical result motivated the binary approach.

Attention helped endo go from 0.000 to 0.225 through spatial focus but did not help ovary at all. Attention also hurt uterus on test (0.557 versus 0.656), probably because the extra SCSE parameters overfit on the small training set.

The binary approach follows the same logic as RAovSeg's slice classifier [5] but with one model instead of two. Training on only organ-containing slices with Focal Tversky Loss (false negatives penalized 4 times more) gave 0.298 for ovary, matching their 0.290 with a simpler setup. The overall mean of 0.389 for binary endometrioma (including non-detected slices) reflects a detection rate of about 70%, which limits clinical deployment but provides a first baseline for this structure.

D. Auto-Prompting

The auto-box results show that localization is the bottleneck for fully automatic MedSAM segmentation. Multi-class U-Net barely finds ovary at all (3 out of 83 slices). Binary models find it more often (63 out of 83) but the resulting Dice of 0.266 is far below the 0.783 from oracle boxes. The decoder was trained only with perfect oracle boxes so it does not handle rough boxes well. Training with perturbed boxes as proposed by Li et al. [16] would directly address this gap, as would the learnable prompt module from Gaillochet et al. [14] which removes the need for boxes entirely.

E. Limitations

I trained everything on CPU which limited U-Net to 15 epochs and meant I could not run nnU-Net which needs GPU for its 3D self-configuring pipeline. I worked with 2D slices and did not use the full 3D volume information. The simulated bounding box uses a formula I derived rather than actual radiologist annotations, though it is grounded in measured inter-rater variability. The decoder still needs oracle boxes at test time. I only report Dice and did not compute boundary metrics like Hausdorff distance which would matter for surgical planning. The binary endo result is from a small test set and would be more reliable with cross-validation.

VI. Future Work

The auto-box pipeline (Section IV-D) showed that the decoder fails with rough boxes because it was trained only with oracle prompts. Training with perturbed boxes [16] would directly address this. The learnable prompt module from Gaillochet et al. [14] could remove the need for boxes entirely by learning to generate prompts from the image embedding.

MedSAM-2 [17] with memory attention could use 3D context across slices, which would help for structures that change shape gradually. Running nnU-Net [9] on GPU would give a strong 3D baseline with automatic class balancing. MedSAM-U [15] could refine prompts using uncertainty estimation.

VII. Conclusion

I evaluated deep learning for pelvic organ segmentation in endometriosis MRI. The data alignment fix (+0.486 Dice) was the single biggest improvement. MedSAM decoder fine-tuning reached 0.857, 0.783, and 0.744 for uterus, ovary, and endometrioma. Binary models matched the published baseline for ovary (0.298 versus 0.290) and gave the first endometrioma result (0.556 when detected). The alignment fix and inter-rater analysis are important for anyone working with this dataset going forward.

References

[1] K. T. Zondervan, C. M. Becker, and S. A. Missmer, "Endometriosis," *N. Engl. J. Med.*, vol. 382, no. 13, pp. 1244–1256, 2020.

[2] C. Chapron et al., "Ovarian endometrioma," *Hum. Reprod.*, vol. 27, no. 3, pp. 702–711, 2012.

[3] S. K. Agarwal et al., "Clinical diagnosis of endometriosis," *Am. J. Obstet. Gynecol.*, vol. 220, no. 4, 2019.

[4] M. Bazot and E. Darai, "Role of MRI in diagnosis of uterine adenomyosis," *Fertil. Steril.*, vol. 109, no. 3, pp. 389–397, 2018.

[5] Y. Liang et al., "UT-EndoMRI: a multi-center pelvic MRI dataset for endometriosis with expert annotations," *Sci. Data*, vol. 12, 2025.

[6] J. Ma et al., "Segment anything in medical images," *Nat. Commun.*, vol. 15, p. 654, 2024.

[7] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. MICCAI*, pp. 234–241, 2015.

[8] O. Oktay et al., "Attention U-Net: Learning where to look for the pancreas," in *Proc. MIDL*, 2018.

[9] F. Isensee et al., "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," *Nat. Methods*, vol. 18, pp. 203–211, 2021.

[10] F. Milletari, N. Navab, and S. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," in *Proc. 3DV*, pp. 565–571, 2016.

[11] T.-Y. Lin et al., "Focal loss for dense object detection," in *Proc. ICCV*, pp. 2980–2988, 2017.

[12] S. S. M. Salehi, D. Erdogmus, and A. Gholipour, "Tversky loss function for image segmentation using 3D fully convolutional deep networks," in *Proc. MLMI*, pp. 379–387, 2017.

[13] A. Kirillov et al., "Segment anything," in *Proc. ICCV*, pp. 4015–4026, 2023.

[14] M. Gaillochet, C. Desrosiers, and H. Lombaert, "Automating MedSAM by learning prompts with weak few-shot supervision," in *Proc. MICCAI-MedAGI*, LNCS 15184, pp. 61–70, 2024.

[15] N. Zhou et al., "MedSAM-U: Uncertainty-guided auto multi-prompt adaptation for reliable MedSAM," *IEEE J. Biomed. Health Inform.*, 2025.

[16] B. Li, Y. Yuan, and W. Tan, "Optimization of MedSAM model based on bounding box adaptive perturbation algorithm," in *Proc. ICIC*, CCIS 2567, 2025.

[17] J. Ma et al., "MedSAM-2: Segment anything in 3D medical images and videos," *arXiv:2504.03600*, 2025.

[18] A. G. Roy, N. Navab, and C. Wachinger, "Concurrent spatial and channel squeeze and excitation in fully convolutional networks," in *Proc. MICCAI*, pp. 421–429, 2018.

[19] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019.

[20] T. Zhao et al., "BiomedParse: a biomedical foundation model for image parsing," *Nat. Methods*, 2024.

[21] Y. Kurata et al., "Automatic segmentation of uterus with malignant tumor on MRI using U-Net," *J. Appl. Clin. Med. Phys.*, vol. 23, no. 7, 2022.